MSG No. 113.274 $P_I\bar{4}2_1m$ [Type IV, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-4'2'm`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[1,4,7,8,10,11,13,14]`
2	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[2,3,5,6,9,12,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `-4'2'm`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[1,4,7,8,10,11,13,14]`
2	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[2,3,5,6,9,12,15,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `22'2'.`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[1,4,10,11]`
2	$\left[0, 0, \frac{1}{4}\right]$	`[2,3,9,12]`
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[5,6,15,16]`
4	$\left[0, 0, \frac{3}{4}\right]$	`[7,8,13,14]`

Table 4: Wyckoff site: `4d`, site symmetry: `-4..`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,4,5,6]`
2	$[0, 0, 0]$	`[2,3,7,8]`
3	$\left[0, 0, \frac{1}{2}\right]$	`[9,12,13,14]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[10,11,15,16]`

Table 5: Wyckoff site: `4e`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	`[1,4,7,8]`
2	$\left[0, \frac{1}{2}, -z\right]$	`[2,3,5,6]`
3	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	`[9,12,15,16]`
4	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	`[10,11,13,14]`

Table 6: Wyckoff site: `8f`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x, 0, \frac{3}{4}\right]$	`[1,10]`
2	$\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[2,9]`
3	$\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{4}\right]$	`[3,12]`

continued ...

Table 6

No.	position	mapping
4	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[4,11]`
5	$\left[0,\ -x,\ \frac{1}{4}\right]$	`[5,15]`
6	$\left[0,\ x,\ \frac{1}{4}\right]$	`[6,16]`
7	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[7,13]`
8	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[8,14]`

Table 7: Wyckoff site: `8g`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,10]`
2	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[2,9]`
3	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[3,12]`
4	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[4,11]`
5	$\left[0,\ -x,\ \frac{3}{4}\right]$	`[5,15]`
6	$\left[0,\ x,\ \frac{3}{4}\right]$	`[6,16]`
7	$\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[7,13]`
8	$\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[8,14]`

Table 8: Wyckoff site: `8h`, site symmetry: `2..`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,4]`
2	$[0,\ 0,\ -z]$	`[2,3]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[5,6]`
4	$[0,\ 0,\ z]$	`[7,8]`
5	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[9,12]`
6	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[10,11]`
7	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[13,14]`
8	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[15,16]`

Table 9: Wyckoff site: `8i`, site symmetry: `..m`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ z\right]$	`[1,8]`
2	$\left[x+\frac{1}{2},\ -x,\ -z\right]$	`[2,5]`
3	$\left[\frac{1}{2}-x,\ x,\ -z\right]$	`[3,6]`
4	$\left[-x,\ \frac{1}{2}-x,\ z\right]$	`[4,7]`
5	$\left[x+\frac{1}{2},\ x,\ z+\frac{1}{2}\right]$	`[9,16]`
6	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	`[10,13]`
7	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[11,14]`
8	$\left[\frac{1}{2}-x,\ -x,\ z+\frac{1}{2}\right]$	`[12,15]`

Table 10: Wyckoff site: `16j`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	`[2]`
3	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$	`[3]`
4	$[-x,\, -y,\, z]$	`[4]`
5	$[y,\, -x,\, -z]$	`[5]`
6	$[-y,\, x,\, -z]$	`[6]`
7	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	`[7]`
8	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z\right]$	`[8]`
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[9]`
10	$\left[x,\, -y,\, \frac{1}{2}-z\right]$	`[10]`
11	$\left[-x,\, y,\, \frac{1}{2}-z\right]$	`[11]`
12	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	`[12]`
13	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	`[13]`
14	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[14]`
15	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	`[15]`
16	$\left[y,\, x,\, z+\frac{1}{2}\right]$	`[16]`